

The traditional fund returns \$110. The 3×Leveraged ETF adds 30% to return \$130. Leveraged ETFs are “reset” daily. Now suppose on day 2, the index falls 10%. The traditional fund drops to $(1 - 0.1) \times 110 = \$99$. The leveraged ETF returns $(1 - 0.3) \times 120 = \$84$. After two days, the index return is $99/100 - 1 = -1\%$. In comparison, the 3×Leverage Fund has returned $84/100 - 1 = -16\%$. This is very different from three times the index return of -1% over two days and contrary to what an ordinary investor might have expected from the fund’s name. Repeat this after many days, and there can be enormous divergence between actual returns and those you expect.⁶⁵

Figure 16.11 plots the S&P 500, the Direxion Daily S&P 500 Bear 3X, and the Direxion Daily S&P 500 Bull 3X. The latter two ETFs are designed to have triple exposure to the S&P 500 in a short and long direction, respectively. Figure 16.11 plots the cumulative return to a \$1 investment in each of the three at the beginning of January 2012 until the end of December 2012. The return of the S&P 500 is -11.8% . The Bear 3X fund returned 74.6% , more than six times the S&P 500 return. The Bull 3X fund returned -31.6% , which is 2.6 times the S&P 500 return. These are very different from the expected “3” in the funds’ names.

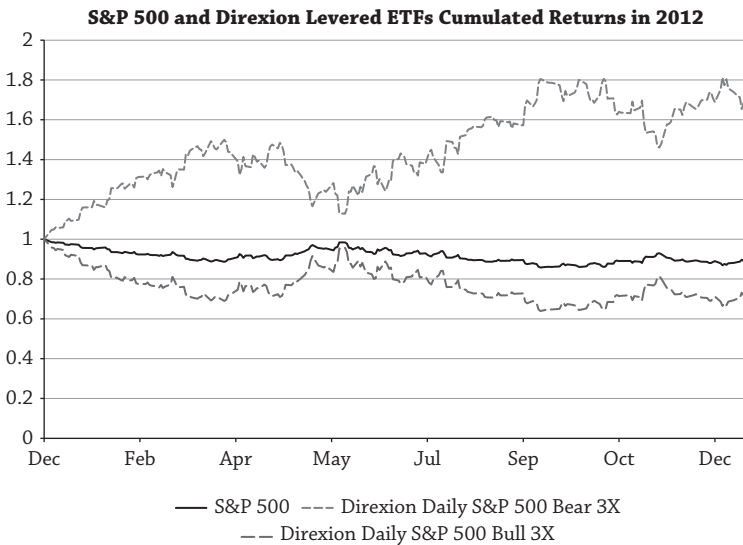


Figure 16.11

⁶⁵ This difference arises from arithmetic vs. geometric returns, see Cheng and Madhavan (2009), chapter 4, and the Appendix. Tang and Xu (2013), however, show that the divergence between the leveraged ETF returns and the leveraged multiple of the underlying index return cannot be explained by just compounding deviations.